Dual-domain Signal Fusion Adversarial Network Based Vision Transformer for Cross-domain Fault Diagnosis

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***Abstract*—Many of the current fault diagnosis methods for intelligence rely on either a signal in the time domain or a signal in the frequency domain. However, frequency-domain signals often fail to fully capture the operational condition of bearings, despite their intuitive nature. On the other hand, time-domain signals contain valuable equipment status information, but their complexity poses challenges for learning by networks. To overcome the challenge, this paper proposes a dual-domain signal fusion adversarial network (DSFAN) based on Vision Transformer. The network is constructed with a parallel feature extractor based on the Vision Transformer. The time domain and frequency domain signal features are integrated considered after parallel extraction and fusion to achieve cross-domain fault diagnosis of bearings. Meanwhile, to enhance the network's ability to extract domain-invariant features, an adversarial training strategy is employed in conjunction with the Wasserstein distance. In order to evaluate the performance of DSFAN, six cross-domain fault diagnosis tasks were conducted for bearing analysis. Compared to other comparison methods, the experimental results demonstrate that the proposed method exhibits superior diagnostic accuracy and domain alignment performance.**

Keywords- fault diagnosis; dual-domain signal collaborative diagnosis; transfer learning; vision transformer

I. Introduction

As rotating machine, the operating status of bearings is of great significance to the safe and stable operation of equipment. The method of implementing bearing fault diagnosis by analyzing vibration signal features has received a wide range of attention [1]. For instant, Zhang et al. [2] developed a fast nonlinear blind deconvolution in order to better extract the bearing vibration signal features in noisy environments. Jiang et al. [3] proposed an adaptive and efficient variational mode decomposition method and applied it to bearing fault diagnosis. An algorithm based on clustering and sparse representation was designed by Lu et al. [4] for identifying faulty bearing signal samples. Meanwhile, with the development of artificial intelligence technology, many scholars have conducted research on deep learning-based fault diagnosis [5]. Shao et al. [6] constructed a dual-threshold attention-guided GAN (DTAGAN) to assist fault diagnosis by generating high-quality samples. Wang et al. [7] proposed a digital twin (DT) aided intelligent fault diagnosis model to overcome the problem of insufficient samples by transferring the DT features. An et al. [8] designed a self-learning transferable neural network (STNN) in order to solve the problem of unlabeled and unbalanced data. In order to

2023 Global Reliability and Prognostics and Health Management Conference (PHM-Hangzhou) | 979-8-3503-0135-9/23/$31.00 ©2023 IEEE | DOI: 10.1109/PHM-HANGZHOU58797.2023.10482802

utilize both simulation and experimental samples, Shao et al. [9] employed a joint transfer network to implement bearing fault diagnosis. An et al. [10] designed a new tool named actively imaginative data augmentation (AIDA) in order to solve the problem of data distribution shifts due to speed fluctuations. A batch normalized sparse filter was proposed by Han et al. [11] to improve the feature distribution in the process of feature extraction. A modified deep subdomain adaptation network (MDSAN) was proposed by Luo et al. [12] to improve fault diagnosis performance by sharing feature extraction modules and a new trade-off factor. These methods contribute to enhancing fault diagnosis accuracy in the respective scenarios. Nonetheless, a majority of these techniques are constrained to fault diagnosis utilizing signals from either the time domain or the frequency domain. The frequency domain signals often do not fully reflect the operating status of bearings, even though the frequency domain signals have good intuition. In addition, the time domain signals have rich equipment status information, but complex the time domain signals are often difficult to be learned by networks. Hence, there is still ample room for improvement in methods for bearing fault diagnosis.

II. THEORETICAL BACKGROUND

A. Unsupervised domain adaptation

Conventional fault diagnosis methods based on deep learning are constrained by their ability to operate solely within specific sample distributions. However, data in engineering practice often exhibit various distributions. Furthermore, the collection of labeled data, which is essential for traditional methods, is often prohibitively costly. To address these challenges, unsupervised domain adaptation methods have been developed. These methods focus on extracting domain-invariant features by minimizing domain discrepancies, enabling data processing across different distributions.

B. Multi-headed self-attention

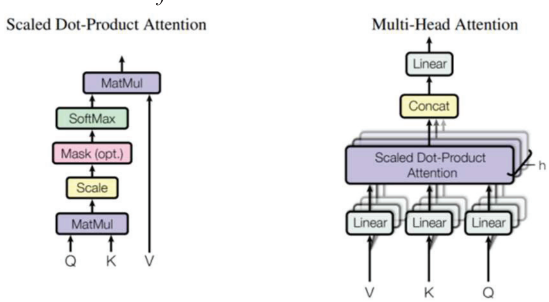


Fig 1. Multi-headed attention mechanism

The multi-headed self-attention (MSA) consists of multiple dot product attention mechanisms. Figure 1 [13] illustrates the schematic diagram of MSA. First, the MSA employs linear mapping to transform the input data into multiple sets of queries, keys, and values. Then, they are pooled separately for attention and stitched together. Finally, the attention pooling matrix is processed by another linear mapping. Attention pooling is calculated as follows:

$$Attention(Q,K,V) = softmax(\frac{QK^T}{\sqrt{d_k}})V \tag{1}$$

where $\sqrt{d_k}$ is the scaling factor.

The process of calculating multiple attention is presented below:

$$h_i = f(W_i^{(q)}q, W_i^{(k)}k, W_i^{(v)}v) \in \mathbb{R}^{Pv} \tag{2}$$

$$Attention = W_0 \begin{bmatrix} h_1 \\ \cdot \\ \cdot \\ \cdot \\ h_h \end{bmatrix} \in \mathbb{R}^{Po} \tag{3}$$

where $q \in \mathbb{R}^{Pv}$、$k \in \mathbb{R}^{Pv}$、$v \in \mathbb{R}^{Pv}$ are the given query, key and value, respectively; each attention head is represented by h_i （$i = 1, \dots, h$）; $W_i^{(q)} \in \mathbb{R}^{Pq \times dq}$, $W_i^{(k)} \in \mathbb{R}^{Pk \times dq}$, *and* $W_i^{(v)} \in \mathbb{R}^{Pv \times dv}$, $W_0 \in \mathbb{R}^{Po \times hqv}$ are learnable parameters.

III. DUAL-DOMAIN SIGNAL FUSION ADVERSARIAL NETWORK (DSFAN) BASED ON VISION TRANSFORMER

DSFAN consists of four main components: a dual-domain feature extractor, a Wasserstein distance metric module, a label classifier, and a domain classifier. Figure 2 displays the framework diagram of the DSFAN. The DSFAN's detailed process unfolds in the following manner:

Step1: The signals from the time domain and frequency domain are separately inputted into parallel feature extractor.

Step2: Dual domain features were fused.

Step3: The fused features undergo the calculation of domain distance using the Wasserstein distance metric.

Step4: The label classifier and the domain classifier receive the fused features for further processing.

To overcome the limitations of relying solely on either time-domain or frequency-domain signal features, we construct a parallel dual-domain signal feature extractor. This allows for the extraction of dual-domain signal features in parallel, resulting in a more comprehensive and representative feature set.

Before inputting the signal into the network, we segment the signal and perform linear projection to obtain input network tokens. For the segmentation and linear projection processes in this paper, convolutional neural network is utilized.

$$z_0 = \text{Linear projection}[x_p^1; x_p^2; \dots; x_p^N] \tag{4}$$

where x is the patch after signal segmentation, z_0 is the tokens. The tokens are overlaid with position embedding and spliced with a learnable class token before being fed into the network.

$$f_0 = [x_{class}; x_p^1E; x_p^2E; x_p^3E; \dots; x_p^NE] + E_{pos} \tag{5}$$

The parallel feature extraction network comprises multiple Vision Transformer Blocks. Each Vision Transformer Block consists of an MSA, two layers of normalization (LN), and a multiple layer perception (MLP). Furthermore, the Vision Transformer Block is equipped with two residual connections.

$$f_l' = MSA(LN(f_{l-1})) + f_{l-1} \tag{6}$$

$$f_l = MLP(LN(f_l')) + f_l' \tag{7}$$

where f_l is the output of the lth Vision Transformer Block.

Finally, the output of the two signal feature extractors is the class token in the features. It is important to highlight that the time domain and frequency domain signal feature extractors are connected to the fusion layer at the end. Within the fusion layer, the class tokens are combined using average pooling. The resulting fused features encapsulate valuable information from both the time and frequency domain signals.

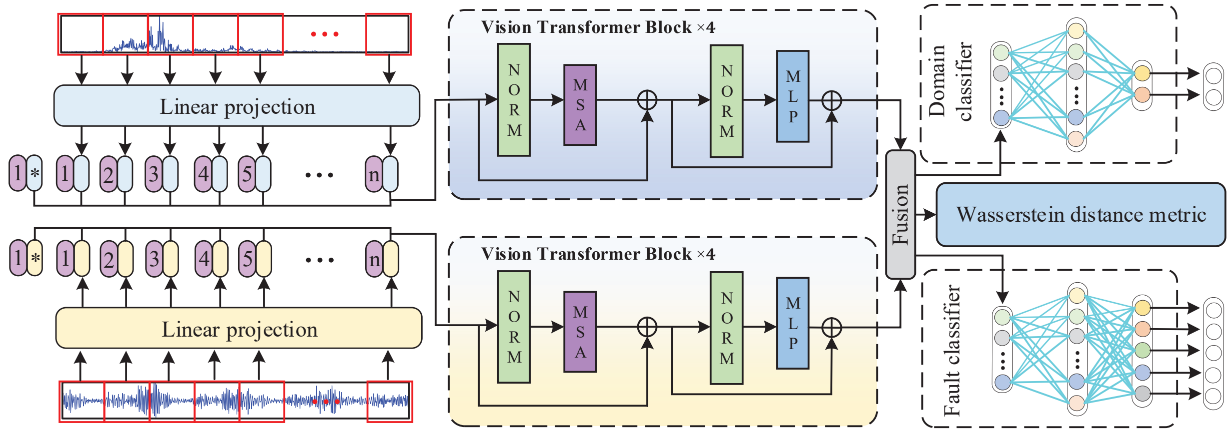


Fig 2. The DSFAN Framework Diagram

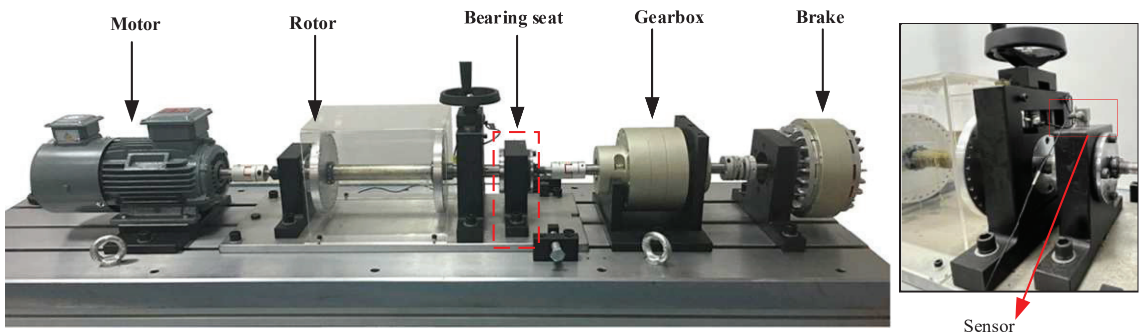


Fig 3. The bearing test bench

TABLE I. FAULTY BEARING INFORMATION

Fault Type	Normal（NC）	Inner ring failure（IF）	Outer ring failure（OF）	Rolling element failure (RF)	Rolling element and outer ring mixing failure (ROF)
Fault Size		0.2	0.2	0.2	0.2

During the training process, the network utilizes the Wasserstein distance to evaluate the fused token, which aids in extracting domain-invariant features. Additionally, the fused token is inputted independently into both the classifier for labels and the classifier for domains. To assess the classification performance, the Softmax cross-entropy is employed. Adversarial training takes place between the two classifiers. This ensures the feature extractor develops a strong capability to extract domain-invariant features.

IV. EXPERIMENTAL VERIFICATION

A. Data description

TABLE II. THREE DIFFERENT WORKING CONDITIONS INFORMATION

Work conditions	1500rpm(A1)	1800rpm(A2)	2000rpm(A3)

The experimental data were gathered by sourcing them from the bearing test bench. Figure 3 displays the photograph of the test stand. The faulty bearing used is the 6205 bearing, intentionally introduced with five faults. Information regarding the faulty bearing is presented in Table I. The experimental configuration encompassed three distinct operational conditions. Six transfer tasks were designed based on these three operational conditions. Tables II and III display information about the operational conditions and transfer tasks. Additionally, the experimental dataset for each faulty bearing consists of 200 samples each in the time and frequency domains. The test and training sets are evenly divided in a 1:1 ratio.

TABLE III. INFORMATION ON THE SIX TRANSFER TASKS

	Task 1	Task 2	Task 3	Task 4	Task 5	Task 6
Transfer tasks	A1 → A2	A2 → A1	A1 → A3	A2 → A3	A3 → A1	A3 → A2

B. Diagnosis results and analysis

TABLE IV. COMPARISON METHODS

Method 1	Transfer component analysis (TCA) [14]
Method 2	Feature-based transfer neural network (FTNN) [15]
Method 3	distance-based deep transfer learning (WD-DTL) [16]

The DSFAN was compared with three traditional transfer learning methods in an experiment. Table IV presents the information regarding the three comparison methods. Furthermore, Figure 4 depicts the diagnostic accuracy diagram for the four methods. It becomes apparent that TCA exhibits a diagnostic accuracy lower than 90% in a majority of the tasks. The diagnostic performance of FTNN is better than that of TCA, and FTNNN can achieve a diagnostic accuracy of more than 90% in most tasks. It is worth noting that WD-DTL has high diagnostic accuracy, with its highest diagnostic accuracy reaching 96.7%. However, the diagnostic accuracy of DSFAN surpasses that of the other three methods in all six tasks.

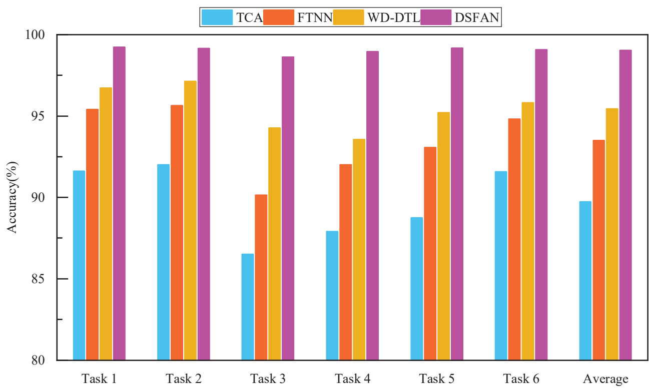
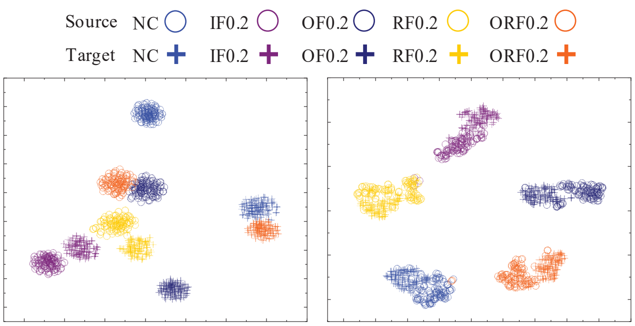
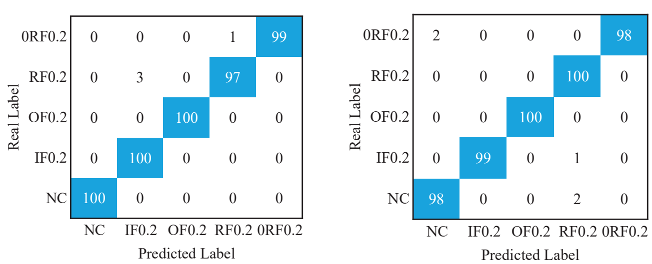


Fig 4. Histogram of diagnostic accuracy

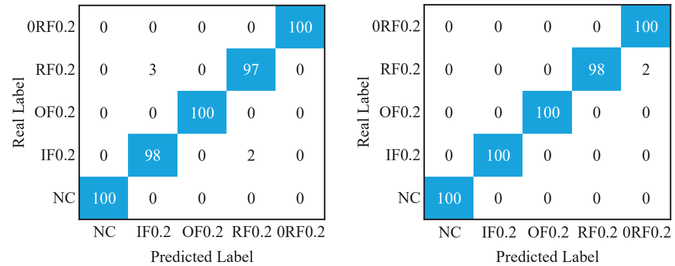


(a) Input Data (b) Output of the feature extractor

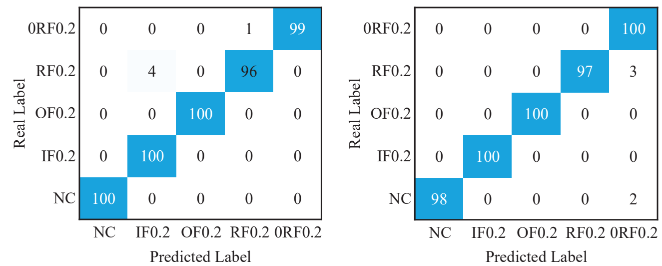
Fig 5. t-SNE dimensionality reduction graph of frequency domain data inputted in network and the output of feature extractor



(a)Task1 (b) Task2



(c)Task3 (d) Task4



(e)Task5 (f) Task6

Fig 6. Confusion matrix for DSFAN diagnosis results

Figure 5 shows the t-SNE [17] dimensionality reduction graph of frequency domain data inputted in network and the output of feature extractor. The graph reveals that the distances between the sample clusters of the input data are very close, with even a slight overlap in some cases. Additionally, the sample clusters in the source and target domains are not aligned. Nevertheless, the parallel network effectively extracts and fuses features, resulting in remarkable domain alignment and distinct separation of sample clusters of different types. The classification confusion matrix of DSFAN in six tasks is presented in Figure 6. The results demonstrate that DSFAN achieves a high level of accuracy in classifying samples of different fault types in each task.

V. Conclusion

In this paper, a dual-domain signal fusion adversarial network (DSFAN) is proposed for obtaining richer information about the operational status of equipment by extracting and fusing time and frequency domain signal features. In addition, transfer learning theory and a parallel feature extraction network constructed by Vision Transformer Block are combined to implement cross-domain fault diagnosis in this paper. In the bearing fault diagnosis experiments, the DSFAN is able to achieve higher diagnostic accuracy compared to other comparative methods and shows strong feature extraction and domain alignment capabilities.

Acknowledgment

The research was supported by the Natural Science Foundation of Shandong Province (ZR2022ME119, ZR202020QE157, ZR2021QE024), the National Natural Science Foundation of China (52105110, 52005303, 52275104), the Natural Science Fund for Excellent Young Scholars of Hunan Province (2021JJ20017) and the Open Project Program of the State Key Laboratory of Mechanical Transmissions of Chongqing University under Grant (SKLMT-MSKFKT-202106).

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